

Self Assessment

MAL2040: Design and Analysis of Experiments

Date: 2026-02-16

Duration: 20 minutes

Total: 30 points

Instructions:

- Do NOT write your name or roll number.
- For each statement, mark Yes / Maybe / No using:
 - Yes: I can do this correctly without notes.
 - Maybe: I could do this with notes or examples.
 - No: I cannot currently do this.
- **Answer honestly.** This is not graded, and it is meant to guide exam preparation.

Lectures 1–2: Introduction to Experimental Design

- Q1. I can describe the three basic principles of experimentation.
- Q2. I can describe the one-factor-at-a-time approach.
- Q3. I know the major drawback of the one-factor-at-a-time approach.
- Q4. I can construct the layout of a 2^2 factorial design.
- Q5. I can compute main effects and interaction effects from a 2^2 factorial design setup.

Lecture 3–4: Probabilistic and Statistical Preliminaries

- Q1. I conceptually understand expectation, variance, covariance, and independence.
- Q2. I can compute sample mean, sample variance, and degrees of freedom from sample data.
- Q3. I know the normal, χ^2 , t , and F distributions.

Lectures 5–6: One-way ANOVA

- Q1. I can write down the fixed-effects one-way ANOVA model with assumptions.
- Q2. I can state the null and alternative hypotheses for equality of means.
- Q3. I can compute SS_T , $SS_{Treatments}$, and SS_E .
- Q4. I can write down the fundamental ANOVA identity and interpret its terms.
- Q5. I can compute the degrees of freedom and the mean squares, $MS_{Treatments}$ and MS_E .
- Q6. I can carry out a complete one-way ANOVA from raw data to final conclusion.

Lectures 7–8: Analysis of the Fixed Effects Model

- Q1.** Given a test statistic or p-value, I can decide whether to reject the null hypothesis.
- Q2.** I know what are Type I errors and Type II errors.
- Q3.** I know what are one-tailed vs. two-tailed tests and critical regions.
- Q4.** I can explain why Cochran's theorem is invoked in the analysis of the fixed effects model.
- Q5.** I know how to compute and interpret a confidence interval.

Lectures 9–11: Model Adequacy Checking

- Q1.** I recall what a residual is.
- Q2.** I know how to make a normal probability plot of residuals and interpret it.
- Q3.** I know how to identify outliers and how to handle them.
- Q4.** I know how to plot residuals vs. run order and interpret the plot.
- Q5.** I know how to plot residuals vs. fitted values and interpret the plot.
- Q6.** I know why balanced designs are preferred.
- Q7.** I know what Bartlett's test does and what its major drawback is.
- Q8.** If ANOVA assumptions fail, I know some possible remedies.

Tutorials on R Programming

- Q1.** I can import a dataset into R, run one-way ANOVA, and obtain the ANOVA table.
- Q2.** I can generate residual plots.
- Q3.** I can run and interpret Bartlett's test.

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- Count of “Yes” answers:
 - Count of “Maybe” answers:
 - Count of “No” answers:

Use this information to assess your exam readiness and to guide your preparation.

End of Self Assessment